

From Number Structure to Particle Selection

A Possible Readout and Its Energy-Pressure Constraints in a Finite WRRR Toy Model

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Abstract

How much microscopic information is needed to refine an aggregate energy model without changing its pressure law? We examine this question through one possible arithmetic-to-particle readout in a finite Worldline-Residue-Resource-Action (WRRR) toy model. A prime-valuation collision statistic assigns conditional probabilities to two supplied lepton-pair labels. This assignment is a modelling assumption, not a physically established interaction or a derivation of particle masses, charges, or spins. An address record provides an isometric extension of the readout; its diagonal probabilities also admit a classical stochastic implementation. We reconstruct the retained phenotype distributions of four archived calibrations and explicitly distinguish this calculation from the inherited resident and return fractions. We characterize when address-dependent energy and its volume derivative can be inferred from aggregate totals. Equal reference energies can erase all address dependence while their derivatives retain it. In a fixed-occupation two-mass example, this prevents an occupied hot channel from preserving a dust-like pressure law. Reversal and constant-routing controls separate consequences of address structure from the chosen species assignment. The result is a reproducible conditional connection and a compatibility constraint for refining the WRRR energy ledger. It is neither empirical confirmation of arithmetic particle selection nor a cosmological lepton-population model.

Keywords: arithmetic address; quantum channel; particle readout; toy model; pressure; identifiability

1 Motivation and the question addressed

The motivating question is whether gravity can be understood as the weight of information. Here that question becomes a precise interface problem: which retained distinctions determine particle observables, and which determine energy and pressure? The r11 model [1] implemented arithmetic admission and retention, a three-sector energy channel, and a conditional homogeneous response. Its aggregate output retains the sector probabilities but erases the address distribution.

The research motivation also has an artistic history. Wonsik Choi describes an early exploration of interference across hypothetical dimensions, followed by an intuition of a finite universe without a boundary while viewing an exhibition by Koo Jeong A. This prompted a shift toward finite integer addresses, prime-factor structure, and explicit filters. The exhibition is a source of a research question, not evidence for the cosmological claim. No actual infinity, infinite-dimensional space, or boundaryless spatial topology is assumed or established by the finite address set used here. The cutoff N specifies this computational instance; it is not an inferred size of the physical universe.

The present extension replaces that loss of resolution with an explicit conditional particle readout. The proposed readout is one possibility within the toy model, rather than a physically verified mechanism. It is a conditional constructive connection from a specified

arithmetic statistic to species probabilities, together with a proof that matching reference energy is insufficient to preserve the original pressure law. This separates a successful interface from an unjustified claim of complete microscopic integration.

2 Input ledger and arithmetic transformation

Verified inputs. We reuse the finite r11 control file supplied with this package: $N = 1,000,000$, exponent $\alpha = 1.8996876950554356$, $K = 8$ frames, and the four archived drive thresholds. Each control had already been calibrated to the same phenotype fraction 0.05, resident fraction 0.268, and return fraction 0.682. We neither refit these thresholds nor infer particle populations from the three totals alone. The two mass anchors are CODATA 2022 central values from NIST [3,4].

Input	Value or role
Electron rest energy ϵ_e	0.51099895069 MeV
Muon rest energy ϵ_μ	105.6583755 MeV
Address cutoff	1,000,000
Reference pair energy E^*	$2 \epsilon_\mu = 211.316751$ MeV
Routing law	Electron-pair probability $\chi(n)$
Volume law	Momentum magnitude proportional to $v^{(-1/3)}$

Verification scope. This package recalculates the phenotype total and its conditional address distribution. The $D = 0.268$ and $R = 0.682$ fractions are inherited r11 inputs to the downstream energy law, not freshly reconstructed filters in this script. The four archived calibrations share those totals in the prior record [1]. Agreement with the known target is a conditional reproduction result; the input/output ledger identifies which part was actually recomputed here.

The uncertainty of ϵ_e is 0.00000000016 MeV; that of ϵ_μ is 0.0000023 MeV. Numerical closure tolerances in this work test arithmetic consistency, not experimental precision. The pair-energy uncertainty is 0.0000046 MeV. No observational likelihood is fitted.

Let O be the odd-composite addresses in $\{2, \dots, N\}$. Write $n = \prod p^{v_p}$ and $\Omega(n) = \sum p^{v_p}$. We introduce the normalized valuation concentration χ . This is an additional, explicit WRRRA readout rule: it treats prime labels symmetrically while retaining multiplicities. It is not claimed to be uniquely required by the r11 model.

$$\chi(n) = \frac{\sum p^{v_p}}{\Omega(n)^2}, 0 < \chi(n) \leq 1. \quad (1)$$

Equivalently, χ is the probability that two independent draws, with replacement, from the multiset of prime-factor occurrences carry the same prime label. This gives a concrete operational meaning to the arithmetic statistic. The rule does not depend on which prime is called first. It is sensitive to repeated use of one source, rather than simply to address magnitude.

Address	Factorization	χ
9	3^2	1
15	3×5	1/2
45	$3^2 \times 5$	5/9
105	$3 \times 5 \times 7$	1/3

For each archived drive d , use weights $w_n = n^{(-\alpha)} / \sum_{m=2 \dots N} m^{(-\alpha)}$, admission $T_{n,d}$, and $\pi_{n,d} = w_n T_{n,d} / \sum_{m \in O} w_m T_{m,d}$. Thus π is conditional on the retained phenotype sector and sums to one. The global probability of a phenotype electron-pair outcome is 0.05 times its conditional probability. Neither probability is a measured cosmological abundance.

For completeness, admission is $T(n,d) = 1 - \text{product over } k = 0, \dots, 7 \text{ of } [1 - \text{sigmoid}(g(n,k,d) - h(d))]$, with $\text{sigmoid}(z) = 1/(1 + \exp(-z))$. For frequency-bearing controls, $g(n,k,d) = J^{(-1/2)} \sum_{j=1, \dots, J} \cos(f(j,d)[\log(n) + \xi(d)k])$, and $J = 4$. For constant_admission, $g = 0$ at every frame. All thresholds below are archived inputs; none is fitted again in this extension.

Control	ξ	Archived h
zeta_dynamic	0.1	1.44767317035244
zeta_static	0	1.3272231882678218
equal_frequency	0.1	1.5258414200781683
constant_admission	not used	1.25537782309341

The two zeta controls use $f = (14.134725141734695, 21.022039638771556, 25.01085758014569, 30.424876125859512)$. The equal-frequency control uses (14, 21, 28, 35). The frame sum and finite address normalization are fully specified; no infinite sum or limiting continuation is needed for the reported result.

A finite computational protocol realizes the chosen rule: prepare the list of $\Omega(n)$ prime-factor occurrences, draw two indices independently and uniformly with replacement, compare their prime labels, and output e on equality and μ otherwise. Counting equal ordered pairs yields sum of νp^2 out of Ω^2 possibilities, exactly $\chi(n)$. This specifies a sampler, not a physical source that produces leptons. Assigning the equality flag to an electron pair is an explicit toy-model convention. Reversing that convention or ignoring the flag gives different allowed readouts.

3 A channel with an explicit address record

Work first on the retained address space with orthonormal basis $|n\rangle$. Let $|e\rangle$ and $|\mu\rangle$ be orthogonal output labels for two-particle states, not single-particle states. Each represents a lepton and its antilepton with opposite momenta and a spin singlet. Their masses, opposite charges, and spin representations are supplied physical identifications. Choosing these representations does not derive charge or spin from integers.

$$|\psi_n\rangle = \sqrt{\chi(n)}|e\rangle + \sqrt{1 - \chi(n)}|\mu\rangle. \quad (2)$$

$$V|n\rangle = |n\rangle_{record} \otimes |\psi_n\rangle, V^\dagger V = I. \quad (3)$$

Proposition 1. Equation (3) is an isometry, even when two addresses have identical χ . Proof. Its matrix elements are $\langle m|V^\dagger V|n\rangle = \delta_{mn} \langle \psi_m|\psi_n\rangle = \delta_{mn}$. Address orthogonality, rather than distinct particle labels, preserves the input identity. The channel $\Phi(\rho) = \text{Tr}_{record}(V\rho V^\dagger)$ is

trace preserving and completely positive [5], with Kraus operators $K_n = |\psi_n\rangle\langle n|$ and $\sum_n K_n^\dagger K_n = I$. Its species probabilities depend only on the diagonal of ρ .

For the r11 diagonal ensemble, the electron-pair probability is $X_d = \sum_n \pi_{n,d} \chi(n)$; the other probability is $1 - X_d$. A further dephasing in the species basis leaves these probabilities unchanged. Therefore the numerical results do not require experimentally accessible coherence between different species. A classical stochastic kernel with $P(e|n) = \chi(n)$ gives the same reported probabilities. The quantum channel supplies a consistent state extension, not evidence of experimentally observed species coherence. The retained record is a mathematical register; its physical realization and energetic cost are not derived here.

$$P_e = \sum_n \chi(n) |n\rangle\langle n|, P_\mu = I - P_e. \quad (4)$$

The reduced two-label output generally cannot reconstruct the original address. In particular, a map that removes $|n\rangle$ record directly from (3) would send distinct orthogonal inputs to nonorthogonal outputs and is not an isometry. The code verifies this negative control. An address-to-particle interface need not be one-to-one: several addresses can legitimately share an observable outcome.

4 What an aggregate can and cannot determine

Let Π_s be the projectors onto the three sectors in the post-filter record space, and $A(\rho) = \sum_s \text{Tr}(\Pi_s \rho) |s\rangle\langle s|$. These projectors refer to the recorded outcomes, not to an assumption that every upstream filter is projective.

Proposition 2. A Hermitian observable O has its expectation recoverable as a linear observable of $A(\rho)$, for every input state, if and only if $O = \sum_s o_s \Pi_s$. **Proof.** The adjoint channel maps every output observable B to $A^*(B) = \sum_s B_{ss} \Pi_s$. Necessity follows because equality of expectations for all density matrices implies equality of operators; sufficiency follows by choosing B diagonal with entries o_s . This is the standard adjoint-channel argument applied to the r11 aggregate, not a claim of a new general channel theorem.

Since $\chi(9) = 1$ and $\chi(15) = 1/2$ within the same sector, the effect P_e is not constant on that sector. Its expectation therefore cannot be recovered universally from the three totals. Conversely, a constant routing rule removes all discrimination among the four controls. Retaining addresses is necessary for this chosen readout but does not establish unique physical identification of those addresses.

5 Particle energy and output-state accounting

Define $\epsilon_a = m_a c^2$. A pair of species a with opposite momenta of magnitude p_a has energy $E_a = 2\sqrt{(\epsilon_a)^2 + (p_a c)^2}$. At a reference volume V_0 , choose the common pair energy $E^* = 2\epsilon_\mu$. The muon pair is then at rest, while the electron pair carries kinetic energy. This is an idealized threshold kinematic state, not a scattering cross section or a production rate.

$$p_\mu c = 0, p_e c = \sqrt{\epsilon_\mu^2 - \epsilon_e^2}. \quad (5)$$

Both pair labels have zero total electric charge and zero net lepton number. We choose the spin singlet $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ in each channel. The code checks that every component of total spin annihilates that state. Net momentum vanishes for the opposite-momentum pair. For an isotropic ensemble, average the pair direction uniformly; this affects the stress tensor interpretation but not the energy expressions.

These are quantum-number-compatible output states. A neutral, spin-zero source with energy E^* can be assigned the same total quantum numbers, but no local interaction Hamiltonian or physical preparation mechanism is derived. The isometry preserves probability, which does not imply energy-preserving dynamics. Output quantum-number closure and rest-plus-kinetic accounting are the checks performed here. No input Hamiltonian or energy-preserving intertwiner is supplied. This is a possible state interface within a toy model, not a demonstrated microscopic conversion process. The broader 15-channel common-carrier program remains background unless its actual operators are inserted into this map.

$$\langle E_{rest} \rangle = 2 \left[X_d \epsilon_e + (1 - X_d) \epsilon_\mu \right], \langle E_{kin} \rangle = E_* - \langle E_{rest} \rangle. \quad (6)$$

Equation (6) partitions one pair-energy budget. The mean rest-energy is a statistical average over known species; it is not the rest mass of a new particle. Adding E^* again to rest and kinetic contributions would double count the same energy. The address probability normalization fixes a fraction, not the absolute number density or physical volume.

Proposition 3. Two pair channels of unequal positive masses cannot both be cold and have the same nonzero per-pair energy with fixed one-pair occupancy. Proof. Cold means $p_a = 0$, so $E_a = 2m_a c^2$. Equal pair energies imply equal masses, contrary to the premise. Different occupancies, a reservoir, or nonzero momenta can relax the premise, but each changes the specified interface and must be entered in the energy ledger.

6 Pressure compatibility under volume change

To test whether the reference-energy equality extends to the r11 dust-like energy law, introduce $v = V/V_0$ and hold channel occupation fixed. Assume freely redshifting momenta $p_a(v) = p_a(1)v^{-1/3}$. This explicitly specified extension is a kinetic diagnostic. Muon decay, collisions, annihilation, and particle production are absent, so it is not a long-time cosmological population model.

$$E_a(v) = 2 \sqrt{\epsilon_a^2 + (p_a(1)c)^2 v^{-2/3}}. \quad (7)$$

$$P_a V = \frac{2 (p_a(1)c)^2 v^{-2/3}}{3 \sqrt{\epsilon_a^2 + (p_a(1)c)^2 v^{-2/3}}}. \quad (8)$$

In (8), P_a denotes pressure, while $p_a(1)$ denotes momentum magnitude. Equation (8) follows directly from $-V \partial E_a / \partial V$. It is nonnegative and is zero exactly for a cold channel. Consequently a mixture with nonnegative fixed weights is dust-like on an interval only if every occupied channel is cold. This is the energy-pressure compatibility obstruction in the present realization.

7 Reproducible numerical case

The code reconstructs all four r11 address distributions from the archived parameters, then applies the same χ rule without fitting it to particle data. The resulting values below are conditional outputs of that declared rule. The displayed precision is chosen to show control differences, not to imply corresponding observational accuracy.

Drive	Electron pair X	Mean rest MeV	Mean kinetic MeV
zeta_dynamic	0.73906640	55.894964	155.421787
zeta_static	0.72871273	58.072287	153.244464
equal_frequency	0.74379972	54.899572	156.417179
constant_admission	0.73992014	55.715428	155.601323

All rows have reference total energy $E^* = 211.316751$ MeV. The newly computed phenotype weights reproduce 0.05 within numerical tolerance; 0.268 and 0.682 remain archived downstream inputs. Nevertheless their species readouts differ. Equation (6) also changes their rest/kinetic allocation. Reversing the association of χ with electron and muon interchanges the species probabilities; the r11 totals cannot determine which association is physical.

Proposition 4. For two normalized retained distributions π and π' , $|X - X'| \leq TV(\pi, \pi')$, where $TV = \frac{1}{2} \sum n |\pi n - \pi' n|$. Proof. Since $0 \leq \chi \leq 1$ and $\sum n (\pi n - \pi' n) = 0$, the positive and negative parts of the signed difference each have mass TV . Multiplying by χ and summing gives the bound. A particle readout can lose distinctions; it cannot amplify total-variation distinguishability beyond this bound. Equal X can arise from distinct address states, so this one readout is not full state identification.

For an explicit comparison with the aggregate ledger, define the dimensionless cell energy $\mathcal{E}(v) = E_{\text{cell}}(v)/E_0$. The normalization assigns 5% of reference cell energy to the pair ensemble; it is an additional allocation assumption. Let $r = (\epsilon_e/\epsilon_\mu)^2$. Keep the r11 D term constant and its R term proportional to volume.

$$E(v) = 0.05 \left[X_d \sqrt{r + (1-r)v^{-2/3}} + 1 - X_d \right] + 0.268 + 0.682 v. \quad (9)$$

$$P(v) = pV/E_0 = -vE'(v), q_{\text{inst}} = \frac{1}{2} \left[1 + \frac{3P(1)}{E(1)} \right]. \quad (10)$$

$$q_{\text{inst}} = -0.523 + 0.025 X_d (1-r). \quad (11)$$

Drive	Phenotype pV/E_0 at $v = 1$	Formal instantaneous q
zeta_dynamic	0.24634971	-0.50452377
zeta_static	0.24289856	-0.50478261
equal_frequency	0.24792744	-0.50440544
constant_admission	0.24663428	-0.50450243

Equation (10) uses the homogeneous acceleration readout employed in r11, with no new geometric derivation. These q values diagnose the supplied equation of state; they are not predictions for the present universe. In particular, replacing its ordinary-matter sector with energetic lepton pairs would change the physical model. Reference energy agreement alone cannot justify retaining $q = -0.523$ after this replacement.

7.1 A criterion for retaining energy and pressure

Proposition 5 (diagonal refinement criterion). On a finite address set partitioned into sectors, let $b(n, v)$ be continuously differentiable in v on an open positive-volume interval, with normalized address weights that are independent of v in each candidate state. Identifiability is quantified over all such normalized address distributions, not just one fixed

distribution. The mean of b is determined by sector totals for every diagonal state and every v if and only if b is constant within each sector at each v . The mean pressure contribution $-v$ times its derivative is determined by those totals if and only if the derivative of b is sectorwise constant. Proof. Concentrating all weight on either of two addresses in one sector proves necessity. Summing a sectorwise constant function proves sufficiency. Apply the same argument to the derivative; v is nonzero. Consequently equal energies at one reference volume do not imply identifiable pressure there. This is an elementary application of aggregate observability to a differentiable energy ledger, not a new general channel theorem.

For any fixed routing function $f(n)$ in $[0,1]$, put $b(n,v) = f(n)E_e(v) + [1 - f(n)]E_\mu(v)$. Within one sector, the difference between two addresses is $[f(n) - f(m)][E_e(v) - E_\mu(v)]$; its derivative has the same form with the energy difference replaced by the derivative difference. Thus a nonconstant f loses energy distinguishability at $v = 1$ when $E_e(1) = E_\mu(1)$, but retains pressure distinguishability whenever their derivatives differ. For the supplied electron/muon energies, $E_e'(1) = -2(\epsilon_\mu^2 - \epsilon_e^2)/(3\epsilon_\mu)$, whereas $E_\mu'(1) = 0$. The witness addresses 9 and 15 therefore have identical reference pair energies but different pressure contributions. A constant f erases both distinctions at every v . This separates an arithmetic choice from the general compatibility condition it must satisfy.

The present χ is nonconstant on the phenotype addresses. Since χ is strictly positive and the electron branch is hot, the chosen realization has positive phenotype pressure. Reversing the rule changes the amount but not the obstruction on these distributions; using a constant positive electron fraction removes control sensitivity but still gives pressure. Restoring dust requires zero occupancy of every hot branch under the declared volume law, or an explicitly changed allocation or dynamics. Neither an alternative numerical fit nor a change of particle names alone removes that constraint.

8 Controls and failure conditions

The executable package runs 47 checks; 47 pass. Tests include reconstruction of the archived 5% weights and address-9 admissions, factor witnesses, isometry, a non-diagonal random-state positivity test, Choi positivity and trace preservation, the failure of naive record removal, spin-singlet closure, the relativistic mass shell, energy allocation, the total-variation bound, pressure finite differences at $v = 0.5, 1$, and 2 , and explicitly executed reversed and constant-routing controls for all four preparations. These are implementation checks, not a count of empirical confirmations.

The derivative test uses a centered step of $10^{-5}v$ and a dimensionless tolerance of 10^{-8} . The random-state check uses seed 10701. The analytic proofs carry the universal statements; the finite numerical tests check the supplied realization. Source files, archived inputs, outputs, and a reproduction README accompany the manuscript.

Test or decision	Failure condition
Address preparation	The phenotype total or address-9 admission is not reconstructed at fixed inputs. D/R are inherited, not tested here.
Quantum interface	Trace, positivity, completeness, or record isometry fails.
Output-state ledger	Assigned pair energy differs from rest plus kinetic energy, or the supplied charge/spin labels fail closure.
Pressure	The derivative of the same volume-dependent energy disagrees with the stated pressure.
Aggregate compatibility	An occupied hot channel is asserted to preserve exact dust pressure without additional terms.
Physical interpretation	A fixed routing and specified preparation disagree with measured species outcomes beyond stated uncertainty.

The physical-interpretation row is a prospective conditional test, not an implemented experimental protocol. A physical preparation mapping measured conditions to n and π , a carrier coupling implementing the routing, and an uncertainty model would have to be fixed before such a comparison could test nature. The implemented failures above test the stated finite toy model. No future physical confirmation is assumed in reporting the present calculation.

9 Scope and relation to earlier work

Earlier WRRR integration work [2] separated component identity, channel selection, address branching, and energy allocation. That distinction is retained here: prime-factor occurrences do not become a count of physical particles, and neither a source register nor a larger channel space multiplies the energy budget. We use the r11 arithmetic controls directly; we do not claim to have executed the earlier full carrier Hamiltonian inside this two-channel example.

The channel sits within the established measure-and-prepare class characterized by Horodecki, Shor and Ruskai [6]; its trace preservation and lack of a need for output coherence are standard facts, not new particle physics. Bény [7] uses coarse-graining channels to study which field-interaction distinctions remain observable at a specified resolution. Our finite model shares the question of retained distinctions, but does not implement that work's field-theoretic metrics or dynamics. Here the task is different and narrower: audit a declared arithmetic refinement of an existing energy ledger against its volume response.

The construction-specific contribution is the chain from finite r11 preparation through a prime-occurrence collision readout to a common pair-energy budget, followed by a derivative compatibility test and explicit alternative-routing controls. Proposition 5 makes the reason for the failure of aggregate pressure identification independent of the particular χ . Its proof and the relativistic kinematics are elementary; the explanatory result is their combined application to the WRRR refinement. These comparisons locate the work among standard tools without asserting exhaustive priority over all arithmetic or informational particle models.

The choice $\chi \rightarrow \text{electron}$ and $1 - \chi \rightarrow \text{muon}$ is a model assumption. A unique mass spectrum, charge assignment, spin representation, interaction law, and lifetime law are not derived. Muon instability is especially important: the frozen-occupation volume derivative cannot be extrapolated to cosmic times. A physical dynamical implementation must include decay products and their energy, charge, and stress, as well as any energetic cost of records. No such additional terms are hidden in the present conservation statement.

These limitations define the scope of the positive result: an address-resolved, mathematically complete conditional readout with a reproducible stress test. The result is stronger than attaching particle names to sector totals because it calculates control-dependent probabilities and identifies a concrete compatibility obstruction. It remains a prototype whose physical routing requires justification or calibration against specified preparation data.

10 Assessment ledger and conclusion

Stage	Result
Verified inputs	Archived r11 parameters and D/R fractions; reconstructed phenotype distributions; known electron/muon rest energies.
WRR transformation	Prime valuation concentration $\rightarrow$ retained address record $\rightarrow$ two pair channels $\rightarrow$ one energy and pressure ledger.
Outputs	Control-dependent species probabilities; exact reference pair-energy closure; a calculated pressure change.
Falsification conditions	Failure of reconstruction, channel completeness, quantum-number closure, energy accounting, pressure differentiation, or a future fixed physical readout.

Within the declared construction, address distinctions survive into particle-channel probabilities even when aggregate sector fractions coincide. This is a completed conditional calculation. The input masses are reproduced by construction, while the arithmetic transformation supplies the channel allocation and its downstream consequences. Neither calibration nor agreement with known values is treated as a defect; the distinction between input and calculated result makes the explanatory claim assessable.

The answer to the opening question is therefore twofold: sector totals suffice for energy only when the energy function is sectorwise constant, and they suffice for pressure only when its volume derivative is sectorwise constant. Equality at one reference point supplies the first property there without supplying the second. This energy-pressure obstruction is the main design constraint for a next implementation. A common reference energy does not guarantee an unchanged gravitational response when the microscopic allocation contains kinetic energy. A viable refinement must either use compatible cold allocations with explicit number normalization, accept the changed pressure, or supply a physically specified compensating sector. Such a sector must carry its own energy and pressure rather than serve as an unrecorded correction.

The present result is an instantaneous kinematic diagnostic of a fixed-occupation toy model that excludes particle decay, production, and annihilation. Application to long-term cosmology requires a dynamical extension that includes decay products and energy transfer. The next physical task is therefore specific: fix the carrier coupling that selects the routing,

identify a preparation and readout procedure, and propagate all occupied modes through one conserved dynamical ledger. This paper supplies the finite interface and the audit needed to assess that task. It does not claim that every address is a unique elementary particle or that all information has a universal mass per bit.

Data and code availability

The accompanying reproducibility archive contains `reproduce.py`, `r11_controls.json`, `results.json`, and this manuscript builder. The earlier r11 record is publicly available at <https://doi.org/10.5281/zenodo.23237629>. This new extension has no assigned public DOI in version 0.2. No new experimental observations were collected.

AI assistance

ChatGPT assisted with mathematical formulation, code, numerical checks, English drafting, and the Korean explanation. These checks are not external journal peer review. Human authors are responsible for evaluating and approving the work before submission.

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